

Virtual Reality Meets Machine Learning: Predicting HRV and Stress Outcomes for Mental Well-Being

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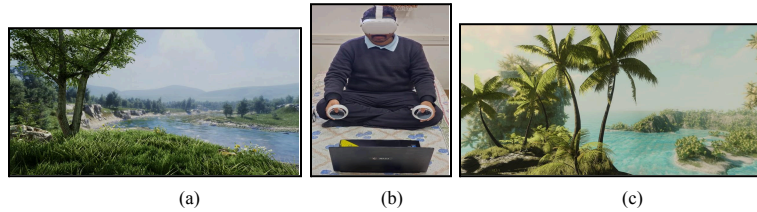


Figure 1: (a) Virtual Reality environment used during the study, (b) Participants practicing Zen meditation within the VR setup (c) A second perspective of the VR environment highlighting its immersive design.

Abstract

Mental health challenges driven by modern stressors necessitate innovative approaches for assessment and intervention. Virtual Reality (VR) provides immersive environments to enhance the accessibility and engagement of meditation practices, while Machine Learning (ML) facilitates precise analysis of physiological markers such as Heart Rate Variability (HRV). This study utilized VR-based meditation and ML techniques to predict changes in objective stress (via HRV) and subjective stress (via depression-anxiety-stress scale), identifying K-Nearest Neighbors as the most accurate model for HRV-based stress prediction (71% accuracy). Logistic Regression demonstrated the highest accuracy for subjective stress change prediction. These findings highlight the potential of VR and ML integration to enable scalable, data-driven mental health solutions.

Introduction

Mental health is a critical global health challenge that affects individuals, communities, and economies worldwide [1]. Modern lifestyles, marked by digital connectivity and societal pressures, have heightened mental health issues, particularly among young people dealing with academic stress, job uncertainties, and the influence of social media [2, 3]. Meditation- and mindfulness-based interventions have proven effective in managing anxiety, depression, and post-traumatic stress disorders [9,10]. Meditation improves cognitive function, reduces mental health symptoms, and enhances well-being [11,12]. While mindfulness practices improve mental health, traditional approaches face barriers to accessibility and engagement. Technologies like Virtual Reality (VR) offer innovative platforms to address these challenges. VR-based meditation has been shown to improve mindfulness and reduce anxiety, especially in academic settings [17]. Systems such as VR-Zen have demonstrated significant improvements in mindfulness and self-reported abilities [20], particularly in novice meditators integrating physical movement and breath awareness [21]. However, evidence

on VR Zen Meditation (VRZM) effectiveness is mixed, with some studies demonstrating benefits in fostering mindfulness and reducing stress, while others report limited impact on psychological and physiological markers [5,6].

Heart Rate Variability (HRV), a robust stress and autonomic balance marker, provides a physiological basis for assessing mental health interventions [7]. Lower HRV reflects higher stress and reduced parasympathetic activity, linking it to emotional regulation and executive functioning [22,23,24]. Although mindfulness techniques improve HRV, emerging approaches like VRZM remain underexplored. The complexity of HRV data necessitates advanced analytical methods to uncover intricate patterns and relationships.

Machine Learning (ML) offers powerful tools for analyzing physiological data like HRV, excelling in identifying patterns and predicting outcomes beyond traditional methods. ML models such as Support Vector Machines, Neural Networks, and Random Forests uncover nonlinear patterns in HRV data [31], providing precise predictions of stress and mental health outcomes [27,28]. While limited, some studies have integrated VR and audio-based meditation with HRV analysis and ML for predictive modeling. Kim et al. demonstrated that VR-based interventions reduce stress and improve mindfulness in immersive environments [1]. Similarly, Marín-Morales et al. highlighted the effectiveness of VR in inducing emotional arousal and mindfulness using heart rate variability as a marker of stress reduction [4]. However, these studies did not explore the integration of audio-based Zen meditation and predictive modeling using ML techniques.

Building upon prior research, this study bridges this gap by uniquely combining VR- and audio-based meditation with HRV analysis and ML algorithms to predict stress and anxiety changes. Unlike earlier works that primarily focused on either VR's immersive effects or ML's analytical capabilities, this study integrates both to provide

a comprehensive framework for mental health interventions. This approach advances ML's application in mental health research and provides insights into the physiological mechanisms of meditation.

MATERIALS AND METHODS

Dataset Description

The dataset used in this investigation was collected from 80 healthy adults aged 18 to 31 years (mean = 24.5 years, SD = 3.2 years). These participants, undergraduate, postgraduate, and PhD students at the Indian Institute of Technology Mandi, were carefully chosen to ensure they had no past mental health disorders or visual impairments. They were randomly allocated to one of four groups:

1. VR-based Zen meditation (VRZM)
2. VR without zen meditation (VR)
3. Audio-only Zen meditation (ZM), and
4. Control group without VR and meditation.

Each group had 20 participants. Figure 1 shows the subject during meditation and the VR environment used in the study. Baseline assessments comprised the Depression, Anxiety, and Stress Scale (DASS-21) and a 5-minute Heart Rate Variability (HRV) recording. Over three days, participants in the intervention groups participated in 10-minute daily sessions according to their group's condition, while the control group went about their normal activities without intervention. Following the third session, post HRV and post-intervention self-reported questionnaire data were collected to examine psychological and physiological changes.

Input Variables

The study aimed to assess the effects of the intervention by separately analyzing both physiological and psychological measures. For the physiological aspect, the study used input variables related to heart rate variability, including Pre-SNS Index which calculates the variation in mean HR about stress, Pre-PNS Index which indicates how much the heart rate changes during exhale compared to inhale, Pre-RMSSD which is the square root of the mean squared variations amongst successive RR intervals, Pre-PNN50 which is pair of RR interval with a succession gap of more than 50 ms, Pre-LF which is peak of low-frequency range (0.04 to 0.15 Hz), and Pre-HF which is peak of the high-frequency range (0.15 to 0.40 Hz), along with the study group classification. These variables were used to examine the impact of the intervention on heart rate variability measures. On the psychological side, self-reported data from the DASS-21 questionnaire, including Pre-Depression, Pre-Anxiety, and Pre-Stress scores, were also included as separate input variables to evaluate the psychological changes before the intervention. These two sets of input variables physiological and

psychological were analyzed independently to better understand their respective effects.

Output Variables

The study analyzed the output variables separately for physiological and psychological measures. For the physiological aspect, the changes in heart rate variability were examined, including change in RMSSD i.e Difference between Post and Pre. On the psychological side, the study also looked at changes in mental health, specifically change in stress, as reported in the DASS-21 questionnaire. Each of these output variables was treated as a distinct classification task to evaluate how the input variables influenced changes in these physiological and psychological metrics..

MACHINE LEARNING ALGORITHMS

To analyze HRV and DASS datasets, we employed machine learning techniques to effectively predict the change in stress levels and other related metrics. The following algorithms were implemented, with hyperparameter tuning carried out to maximize performance. Additionally, we used a train-test split of 7:3 along with 5-fold cross-validation to ensure robust evaluation and generalizability of the models.

Logistic Regression (LR) models probabilities using the logistic function and is effective in HRV-based stress detection. Decision Trees (DT) are non-parametric algorithms that recursively split data, enabling complex interaction predictions. Random Forest (RF), an ensemble of decision trees, enhances accuracy and robustness in HRV classification. Support Vector Classifier (SVC) constructs hyperplanes to separate classes, using kernels to handle nonlinear relationships, excelling in stress classification. K-Nearest Neighbors (KNN) classifies data points by neighbor similarity, making it suitable for exploratory HRV analysis, with hyperparameter tuning boosting prediction accuracy[1,2]

Model Architecture

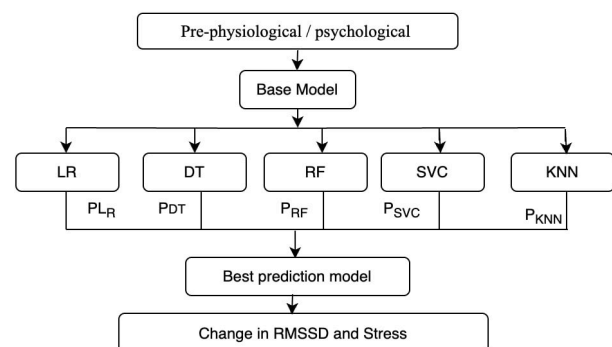


Fig 2. ML architecture for Predicting

Performance assessment matrices

Accuracy measures the proportion of correct predictions made by the model, calculated as the ratio of correctly predicted changes in RMSSD or mental health indicators to the total predictions. Precision evaluates how well the model avoids false positives, representing the ratio of correctly predicted positive changes in RMSSD and DASS variables to all predicted positives. Recall assesses the model's ability to identify positive changes, defined as the ratio of correctly identified positives to the total actual positives. The F1-score, the harmonic mean of precision and recall, balances precision and recall for predicting changes in RMSSD and DASS indicators.

Model calibration

To optimize model performance, hyperparameter tuning was conducted using grid search with 5-fold cross-validation. Each model was calibrated by systematically varying key hyperparameters, ensuring optimal accuracy and robustness. The tuned parameters for each algorithm are outlined in the table below, reflecting a balance between model complexity and generalization. This comprehensive calibration ensures improved prediction quality for the classification task.

Model	Hyperparameter	Range/Options
LR	Regularization strength	0.01 to 5.0 (step 0.5)
	Penalty	l1, l2
DT	Max tree depth	None, 5 to 100 (step 5)
	Min samples to split node	2 to 50 (step 2)
	Min samples in a leaf	1 to 50 (step 2)
RF	Number of estimators	50 to 1000 (step 100)
	Max tree depth	None, 5 to 100 (step 10)
	Max features to split node	sqrt, log2
SVC	Penalty	0.01 to 5.0 (step 0.5)
	Kernel	linear, rbf, poly
KNN	Number of neighbors	3 to 50 (step 2)
	Weights	uniform, distance

Table 1: Hyperparameter Tuning and Calibration Parameters

Result

Change in RMSSD

Machine learning models varied in performance for RMSSD prediction. KNN achieved the highest testing accuracy (0.71), followed by DT and SVC (0.67). Random Forest overfitted with perfect training accuracy but lower test accuracy (0.62). Logistic Regression showed the lowest performance (0.58), highlighting KNN algorithm strong generalizability.

Model	Metric	Train	Test
LR	Accuracy	0.65	0.58
	Precision	0.65	0.63
	Recall	0.65	0.58
	F1-Score	0.65	0.57
DT	Accuracy	0.85	0.67
	Precision	0.85	0.68
	Recall	0.85	0.67
	F1-Score	0.85	0.67
RF	Accuracy	1.00	0.62
	Precision	1.00	0.64
	Recall	1.00	0.62
	F1-Score	1.00	0.62
SVC	Accuracy	0.78	0.67
	Precision	0.78	0.70
	Recall	0.77	0.67
	F1-Score	0.78	0.66
KNN	Accuracy	0.65	0.71
	Precision	0.65	0.71
	Recall	0.65	0.71
	F1-Score	0.65	0.71

Table 2: Performance Comparison of Machine Learning Models for Change in RMSSD Prediction.

Change in Stress

Model	Metric	Train	Test
LR	Accuracy	0.66	0.68
	Precision	0.67	0.70
	Recall	0.66	0.68
	F1-Score	0.66	0.67
DT	Accuracy	0.66	0.64
	Precision	0.70	0.71
	Recall	0.66	0.64
	F1-Score	0.65	0.61
RF	Accuracy	1.00	0.60
	Precision	1.00	0.60
	Recall	1.00	0.60
	F1-Score	1.00	0.60
SVC	Accuracy	0.71	0.60
	Precision	0.77	0.61
	Recall	0.71	0.60
	F1-Score	0.69	0.59
KNN	Accuracy	0.64	0.64
	Precision	0.64	0.64
	Recall	0.64	0.64
	F1-Score	0.63	0.64

Table 3: Performance Comparison of Machine Learning Models for Change in Stress Prediction.

Stress change prediction showed LR as the most effective model with a test accuracy of 0.68. DT, KNN, and SVC achieved moderate accuracies (0.60 -- 0.64). Random

Forest overfitted with perfect training accuracy but poor testing performance (0.60). Logistic Regression demonstrated strong predictive reliability for stress level changes.

Overall, KNN excelled in predicting RMSSD changes, while Logistic Regression was the best for stress predictions. These results affirm machine learning's utility in analyzing physiological and psychological responses.

Discussion and Conclusion

This study highlights the transformative potential of Virtual Reality Zen Meditation (VRZM) combined with machine learning (ML) analytics for enhancing mental and physiological well-being. The VRZM group demonstrated significant improvements in Heart Rate Variability (HRV) parameters, such as RMSSD indices, which are robust markers of autonomic nervous system regulation. These findings align with research by Kim et al. (2021) and Marin-Morales et al. (2021), which reported VR's ability to foster relaxation and mindfulness through immersive environments. Similarly, the results are consistent with studies such as Navarro-Haro et al. (2017), emphasizing that VR enhances emotional regulation by engaging multiple senses.

The integration of machine learning further amplified the study's analytical power, enabling precise predictions of physiological and psychological changes. K-Nearest Neighbors (KNN) achieved the highest test accuracy (71%) for RMSSD change prediction, while Logistic Regression showed strong accuracy for stress change prediction (68%). These outcomes corroborate findings by Razavi et al. (2023), who demonstrated ML's efficacy in identifying subtle physiological changes related to stress. The use of ML in HRV-based research has also been highlighted by Holzman and Bridgett (2017), who emphasized the predictive accuracy of Support Vector Machines (SVMs) and Random Forest (RF) for stress and emotional state detection.

Additionally, studies by Cho et al. (2020) demonstrated ML's ability to handle high-dimensional physiological datasets, facilitating better predictions of mental health outcomes. Our study echoes this capability, with KNN and Logistic Regression effectively modeling the nonlinear relationships in HRV data. However, Random Forest's overfitting challenges in this study align with Farhan et al. (2018), who noted the need for balanced hyperparameter tuning to improve generalizability in ML-based mental health research scores on the DASS-21 scale, consistent with findings by Chandrasiri et al. (2022) and Feinberg et al. (2022). These studies similarly attributed the enhanced outcomes to VR's immersive qualities, which sustain engagement and focus during mindfulness sessions. The limited improvements observed in the VR-only group reaffirm the necessity of guided mindfulness techniques for meaningful results.

Psychological outcomes further substantiate VRZM's efficacy. Participants in the VRZM group reported

significant reductions in depression, anxiety, and stress. The integration of VR and ML in this study supports broader trends in mental health research, where technology is increasingly leveraged to augment traditional interventions. The findings align with research by Razavi et al. (2022), who demonstrated how ML could enhance replicability and scalability in mindfulness-based studies. Together, VR and ML provide a robust framework for precise, data-driven mental health interventions.

This study validates the efficacy of Virtual Reality Zen Meditation (VRZM) as an innovative tool for stress reduction and mental well-being. The results highlight VRZM's ability to significantly enhance HRV metrics and reduce psychological distress, consistent with prior research by Kim et al. (2021), Navarro-Haro et al. (2017), and Marin-Morales et al. (2021). The immersive nature of VR, when coupled with guided meditation, enhances relaxation and user engagement, distinguishing VRZM from traditional mindfulness practices.

The incorporation of machine learning further underscores its potential in mental health research. With KNN achieving the highest predictive accuracy for HRV changes and Logistic Regression excelling in stress prediction, this study aligns with findings by Razavi et al. (2023) and Orrú et al. (2020). The ability of ML to analyze complex physiological data reinforces its utility in creating scalable, personalized interventions.

In conclusion, VRZM, augmented by machine learning, represents a scalable and scientifically grounded approach to addressing modern mental health challenges. By bridging immersive VR experiences with predictive analytics, this study contributes to the growing body of evidence supporting technology-driven mental health solutions and sets the stage for future innovations in this field.

This study was limited by the small sample size of 80 healthy participants aged 18–31 years, which restricts its generalizability. Excluding individuals with mental health conditions limits insights into their potential benefits. The narrow age range also precludes broader applicability. Future work should include diverse age groups and participants with mental health challenges to improve findings. Additionally, addressing Random Forest model overfitting through better validation techniques, such as regularization or ensemble methods, is recommended to enhance testing performance and generalizability.

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